

30	D	Mass difference $\Delta m = 4.002603u + 8.005305u - 12.000000u$ $= 0.007908u$ Energy given off in the nuclear fusion reaction $E = \Delta mc^2$ $= (0.007908)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$ $= 1.1815 \times 10^{-12} \text{ J}$ $= \frac{1.1815 \times 10^{-12}}{1.60 \times 10^{-19}}$ $= 7.384 \times 10^6 \text{ eV}$ $\approx 7.38 \text{ MeV}$